

Hena Kharwa

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Experience

Stevens Institute of Technology

Apr 2025 – May 2026

Research Assistant

Hoboken, New Jersey, United States

- Designed and evaluated RAG-based anomaly detector on 10K compliance docs; 96% precision vs 81% BERT baseline
- Reduced hallucination rate to 10% on compliance queries via LoRA fine-tuning on Llama-3 8B with RAG pipeline
- Architected serverless Lambda + S3 + EC2 ingestion pipeline processing 500K+ docs/month at sub-200ms p95 inference latency
- Boosted retrieval precision by 22% over dense-only baseline by fusing KG entity links with FAISS/Pinecone hybrid search

Enlighten Infosystems

Nov 2025 – May 2026

Generative AI Engineer

Vadodara, Gujarat, India

- Developed modular GenAI pipelines using LangChain + LangGraph, reducing multi-step LLM task latency by 35% in production
- Fine-tuned GPT/LLaMA models with PEFT/LoRA on domain-specific corpora, boosting eval task accuracy by 28% in production
- Built production RAG systems with FAISS/Pinecone + HuggingFace embeddings, improving semantic retrieval precision by 30%
- Deployed scalable GenAI APIs via FastAPI + Docker on AWS Lambda, cutting end-to-end model inference response time by 25%

Dhyey Technologies - TechXi

Dec 2023 – July 2024

AI/ML Engineer

Vadodara, Gujarat, India

- Led GPU optimization of production DL models; cut training time 12→8.4 hrs via mixed-precision TF, accuracy 58%→81%
- Fine-tuned BERT/GPT-2 on 50K-doc domain corpus for summarization; improved retrieval MRR@10 by 25% vs TF-IDF baseline
- Cut real-time inference latency by 35% migrating batch pipelines to async FastAPI on Azure Containers (2K+ req/day)
- Compressed model deployment cycle from 3 days → 5 hrs via GitHub Actions + Azure ML CI/CD pipeline with ONNX export

AtliQ Technologies

Sep 2022 – Nov 2023

AI/ML Developer

Vadodara, Gujarat, India

- Owned end-to-end hybrid CF + content-based recommender in PyTorch/CUDA; cut misclassification by 25% (200K+ catalog)
- Replaced rule-based recs with BERT + Word2Vec semantic embeddings; lifted CTR 20%, conversion 5% in A/B test (100K MAU)
- Indexed 500K+ product embeddings in FAISS using DistilBERT; improved top-5 suggestion accuracy by 18% vs keyword search
- Containerized rec microservices with Docker + FastAPI + Azure ML serving 50K+ daily requests at p99 150ms latency (25%)

Enlighten Infosystems

May 2022 – Aug 2022

Python and ML Intern

Vadodara, Gujarat, India

- Implemented expert system in PyTorch/CUDA for decision classification; cut avg decision time 4.5→3.1 min, +10% accuracy
- Built ResNet-18 + OpenCV visual inspection pipeline; lifted accuracy 15% vs threshold baseline, 800ms → 600ms per image
- Shipped Streamlit dashboards with live real-time ML predictions for 15 stakeholders; 90% satisfaction score (n=20 survey)
- Prototyped BERT auto-summarizer for internal compliance reports on 5K docs; cut generation time 2 hrs → 96 min, +15% F1

Projects

Hospitalist Companion App | 1st Place - Rutgers Health Hack 2025 (RWJUH Spezi App Track)

Oct 2025

- Built an AI-powered clinical iOS app with SwiftUI, Stanford's Spezi framework, and GPT-5, boosting physician query speed 2x.
- Integrated a GPT-5/AWS Bedrock RAG pipeline with FHIR/HIPAA-secure backend and semantic search, lifting accuracy by 35%.
- Enabled hands-free voice queries via AssemblyAI/AVSpeechSynthesizer on a Spezi, S3+Firebase, HIPAA/RBAC backend.

AI-Driven Personalized Alumni Engagement System | Azure AI Foundry, LangChain, React, Docker

Apr 2025

- Built Python/Quart + Node.js backend with FAISS vector embeddings, achieving 25% match precision across 50K+ profiles
- Applied LangChain + Azure AI Foundry for LLM orchestration, enabling personalized multi-turn alumni-student recommendations
- Integrated React web app on Azure serverless infrastructure with Docker + GitHub CI/CD, supporting real-time engagement

PrepRight: AI Powered Interview Prep App | Python, React, Flask, BERT, FAISS, HuggingFace

Jan 2025

- Created interactive dashboards/resume parsers/feedback modules with React.js + Flask APIs and interview assessments on AWS
- Fine-tuned BERT/GPT models via Hugging Face on domain data, boosting accuracy by 25% and deploying low-latency APIs
- Vectorized 1K+ resumes and JDs using FAISS embeddings, achieving 25% improvement in semantic matching accuracy overall

Technical Skills

Languages: Python, Java, C++, Swift, SQL, JavaScript, TypeScript Shell Scripting

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, JAX, TensorRT, TensorRT-LLM, VLLM, LORA, FSDP

Agentic AI & LLM Orchestration: LangChain, LangGraph, LangSmith, Crew AI, AutoGen 0.4, RAG, Knowledge Graph, MCP

Deployment & Infrastructure: FastAPI, Flask, Docker, Kubernetes, FAISS, Pinecone, PostgreSQL, Github Actions, Git, CI/CD

AWS & Cloud AI: AWS(EC2, SageMaker, S3, Lambda, Bedrock), GCP, Azure AI Foundry, Neo4j, PostgreSQL, Redis, Pinecone

Foundation Models: Claude, OpenAI GPT, LLaMA, HuggingFace, BERT

Education

Stevens Institute of Technology

Sep 2024 – May 2026

Masters of Science in Applied Artificial Intelligence | GPA: 4.0/4.0

Hoboken, New Jersey, USA

Charotar University of Science and Technology

Oct 2020 – May 2024

Bachelors of Technology in Computer Engineering | GPA: 3.84/4.0

Anand, Gujarat, India

References

Min Song | Director, Center for Innovative Computing @ Stevens | msong6@stevens.edu | +1 (201) 216-5291

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